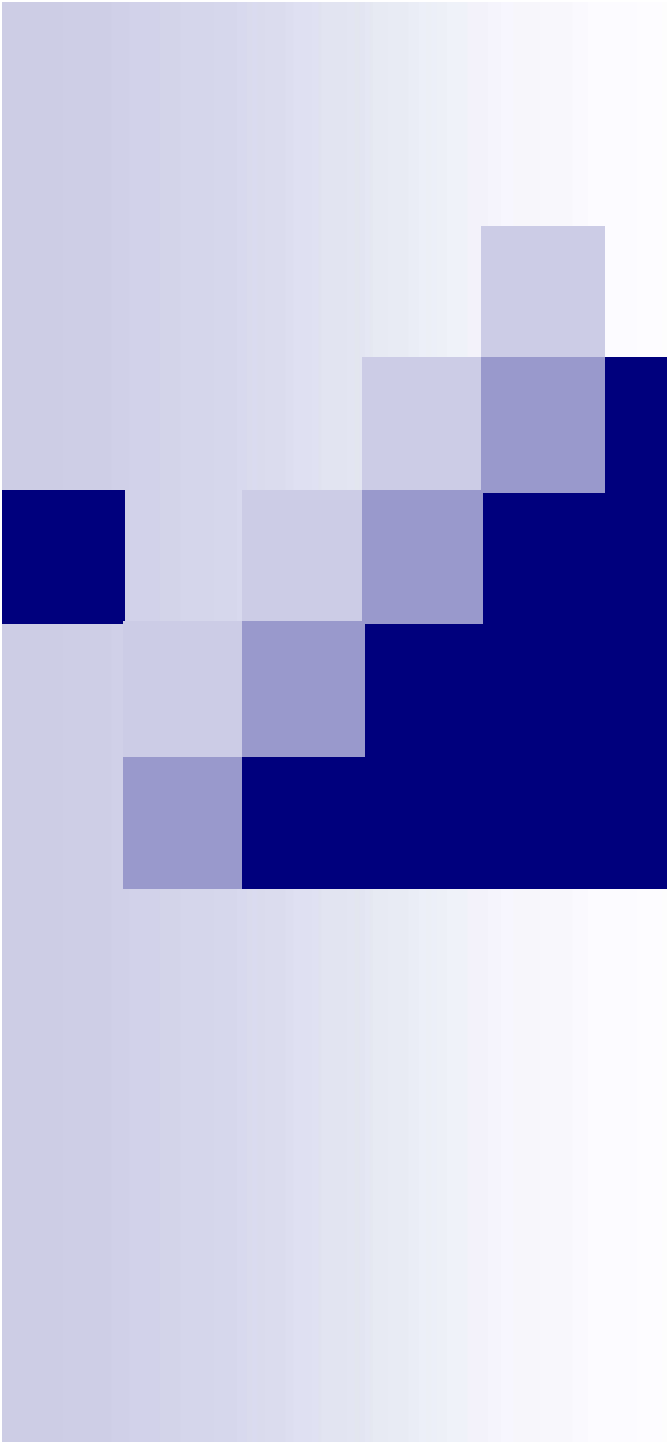




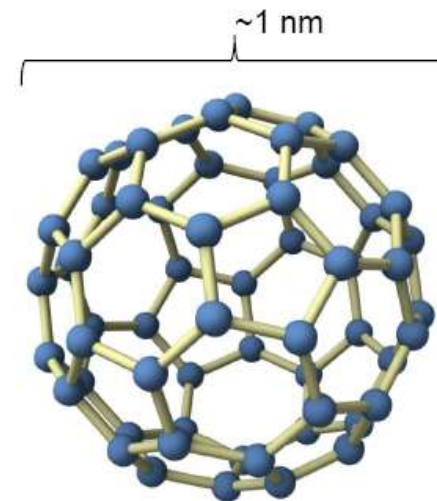
INTRODUCTION TO NANOMATERIALS

What is Nanoscale

- Nanoscale : generally refers to the size scale of 1 – 100 nm in at least one dimension.

1 nanometer (nm) = 10^{-9} m

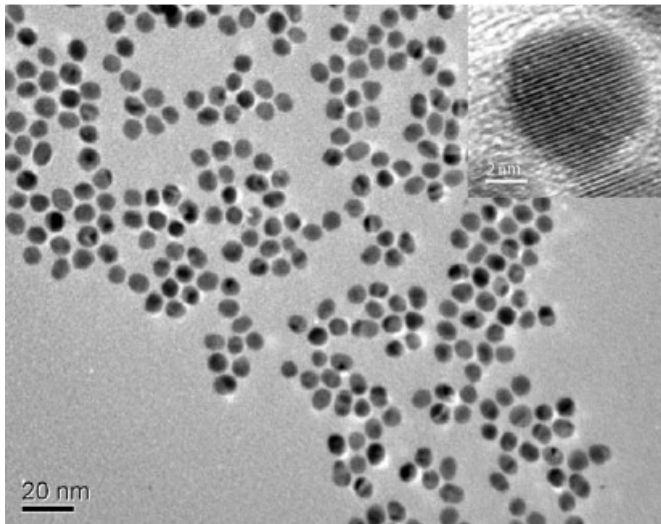
An example is a bucky ball
- a fullerene



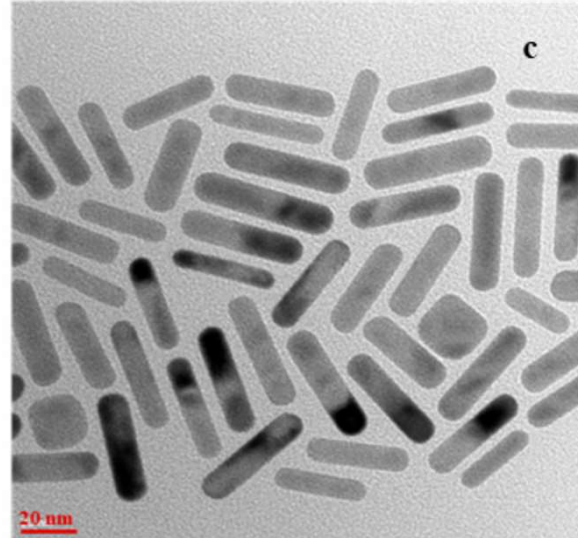
- 60 carbon atoms linked together in one unit.

What are Nanomaterials?

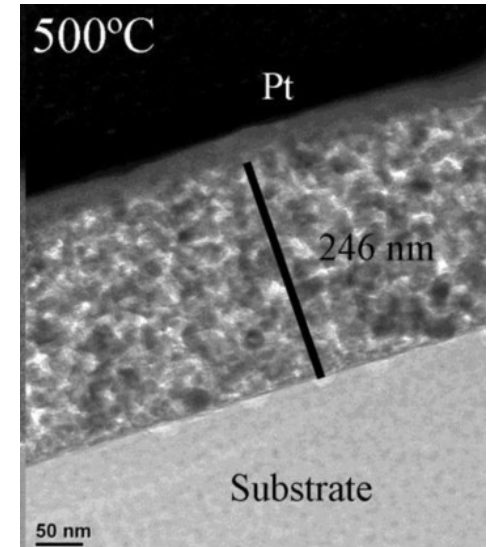
A material that has at least one dimension (height, width or length) that is smaller than 100 nanometres.



Ag NPs

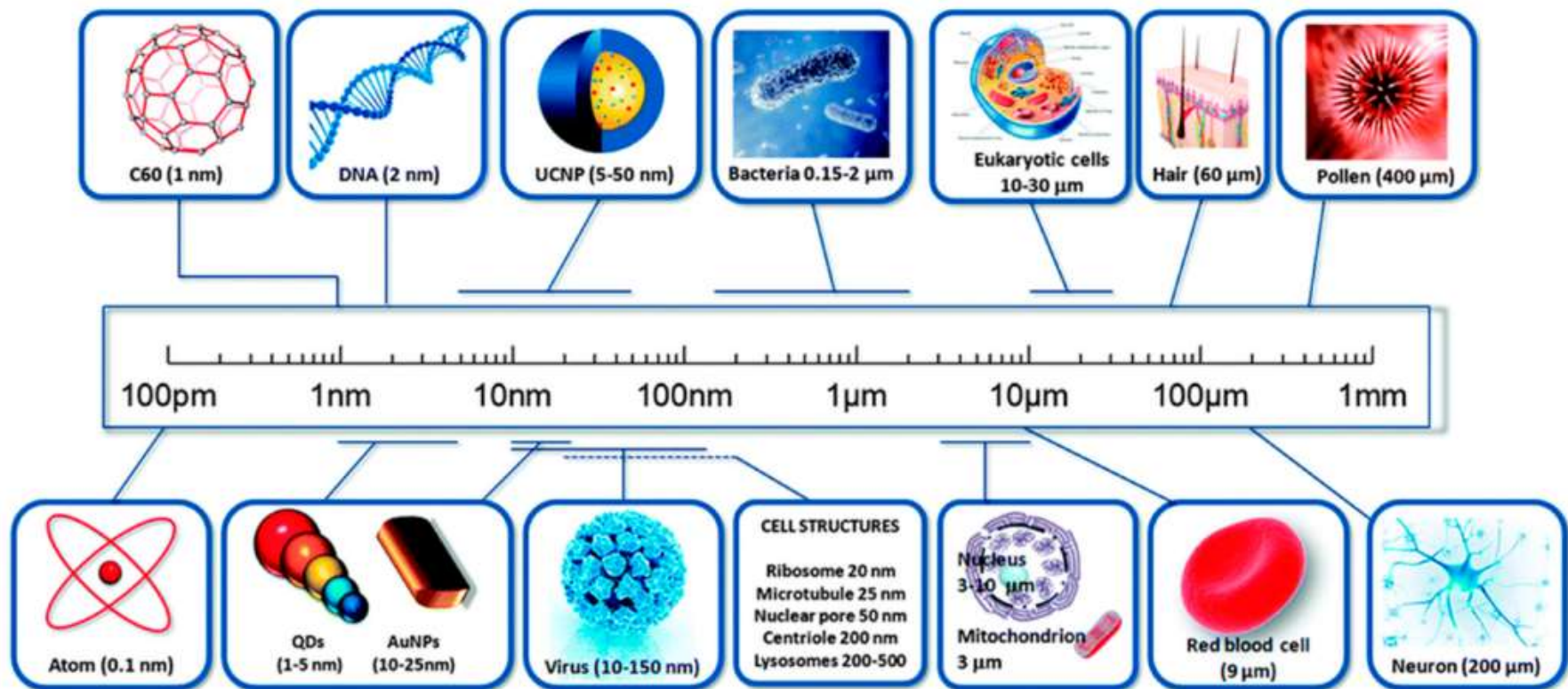


Au Nanorods



TiO₂ Thin Film

A comparison of sizes of nanomaterial.

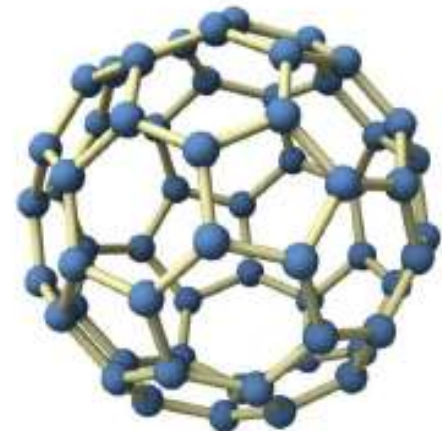


How small is Nano?

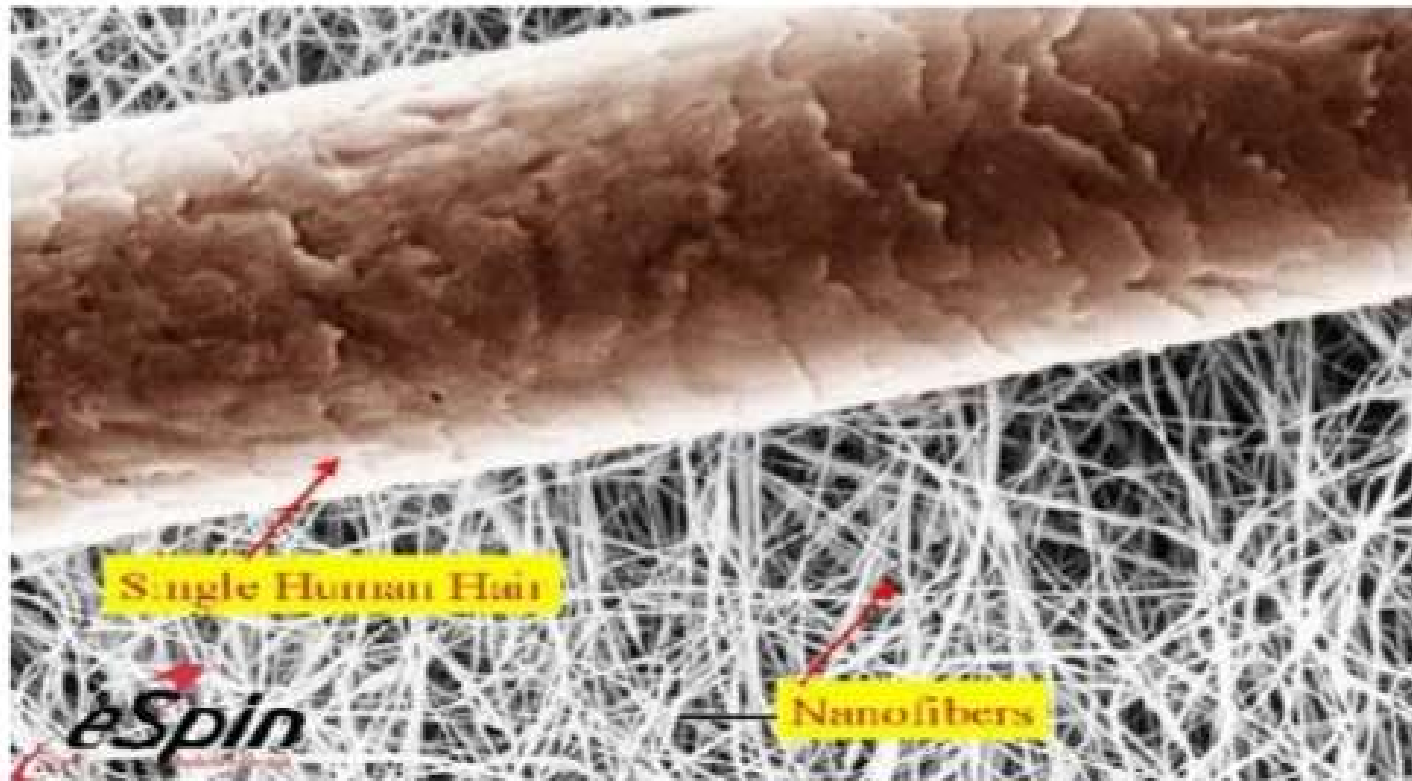
ratio
earth / football = ratio
football / fullerene



12,800 km

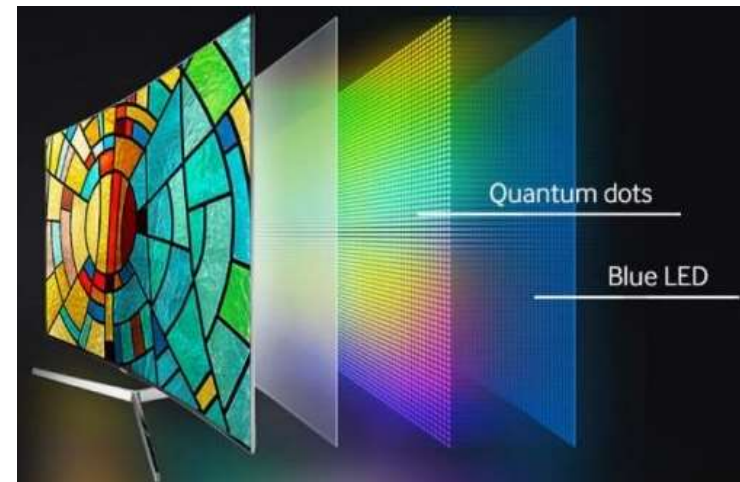


- The head of a **pin** is about 1 million nanometers across.
- A **human hair** is about 60,000 nanometers in diameter, and a **DNA** molecule is between 2-12 nanometers wide.



Nanoscience & Nanotechnology

- **Nanoscience**-is the study of nanomaterials, their properties and related phenomena.
- **Nanotechnology**-is the application of nanoscience to produce devices and products.





Why Nanotechnology?

1. Thermal Properties:

1. The **melting point** of materials can shift due to their reduced size at the nanoscale.
2. Heat transfer behavior may differ significantly.

2. Optical Properties:

1. **Fluorescence**, **color**, and other optical characteristics can change.
2. Nanoscale particles exhibit unique interactions with light.

3. Electrical Properties:

1. **Conductivity** can be enhanced or modified.
2. Quantum effects play a crucial role in electrical behavior.

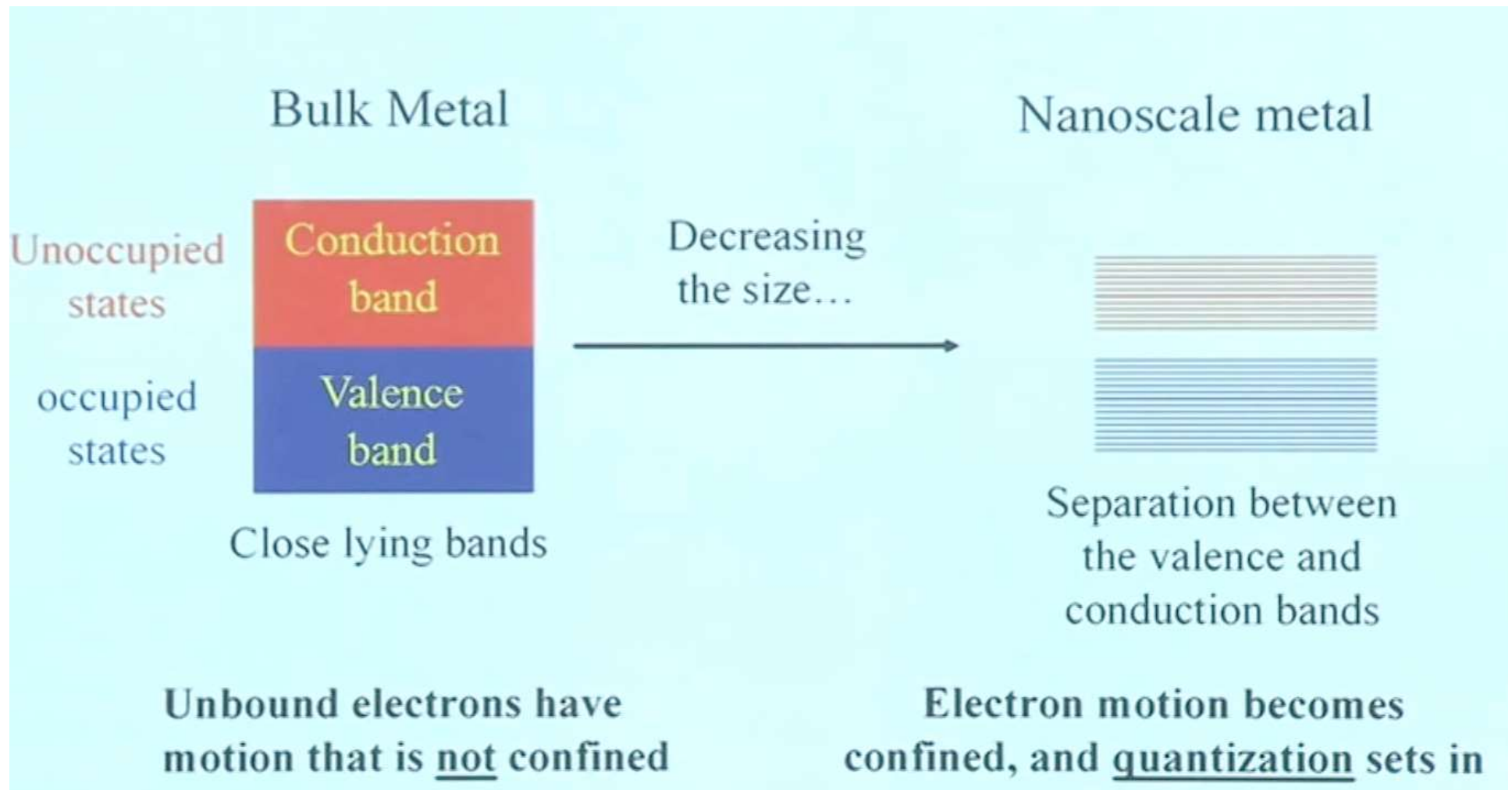
4. Magnetic Properties:

1. **Permeability** and other magnetic traits may vary.
2. Nanoscale magnetic materials find applications in data storage and sensors.

5. Chemical Properties:

1. **Reactivity** alters due to increased surface area-to-volume ratio.
2. Nanoparticles can act as efficient **catalysts**.

Origin of the properties





SIZE DEPENDENT PROPERTIES OF NANOMATERIALS

- Physical Properties (e.g. melting point)
- Chemical Properties (e.g. reactivity, catalysis)
- Electrical Properties (e.g. conductivity)
- Mechanical Properties (e.g. strength)
- Optical Properties (e.g. Absorption and scattering of light)
- Magnetic properties: Magnetization

Material properties vary with size of material

Atoms:
colorless, 1 Å

Gold clusters:
orange, nonmetallic,
<1 nm

Gold nanoparticles:
3–30 nm, red, metallic,
“transparent”

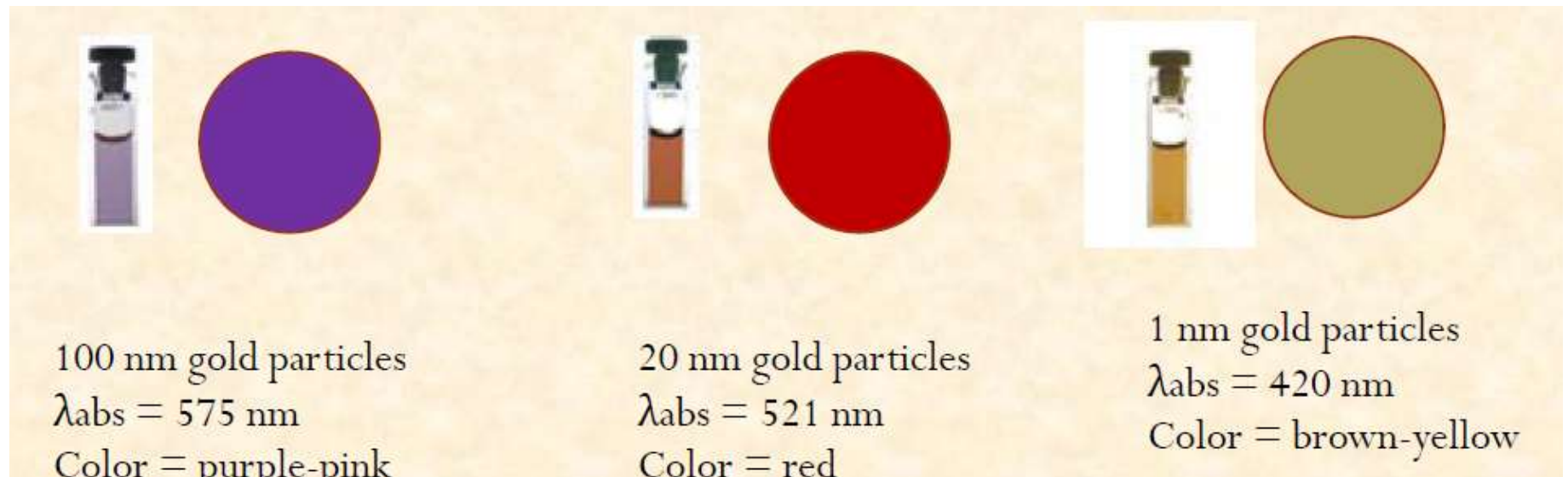
Gold particles:
30–500 nm
metallic, turbid,
crimson to blue

Bulk gold film



- (Bulk) Gold is a shiny yellow metal
- Gold (Au) nanoparticles appears red
- Bulk gold does not exhibit catalytic properties
- Au nanoparticle is an excellent low temperature catalyst

What is a colour of Gold?



Reduction of particle size from micron to nanometer scale influences their optical properties.

Different sizes of particles reflect and absorb light differently.



Physical Properties of Nanomaterials

1. **Reduced Melting Point** -- Nanomaterials may have a significantly lower melting point or phase transition temperature and appreciably **reduced lattice constants (spacing between atoms is reduced)**, **due to a huge fraction of surface atoms** in the total amount of atoms.
2. **Ultra Hard** -- Mechanical properties of nanomaterials may reach the theoretical strength, which are one or two orders of magnitude higher than that of single crystals in the bulk form. The enhancement in mechanical strength is simply **due to the reduced probability of defects**.
3. **Optical properties** of nanomaterials can be significantly different from bulk crystals.
 - **Semiconductor Blue Shift** in adsorption and emission **due to an increased band gap**.
Quantum Size Effects, Particle in a box.
 - **Metallic Nanoparticles Color Changes** in spectra **due to Surface Plasmons Resonances**
Lorentz Oscillator Model.
4. **Electrical conductivity decreases** with a reduced dimension **due to increased surface scattering**.
Electrical conductivity increases due to the **better ordering** and **ballistic transport**.
5. **Magnetic properties** of nanostructured materials are distinctly different from that of bulk materials.
Ferromagnetism disappears and transfers to **superparamagnetism** in the nanometer scale **due to the huge surface energy**.
6. **Self-purification** is an intrinsic thermodynamic property of nanostructures and nanomaterials **due to enhanced diffusion of impurities/defects/dislocations** to the nearby surface.
7. **Increased perfection enhances chemical stability**.

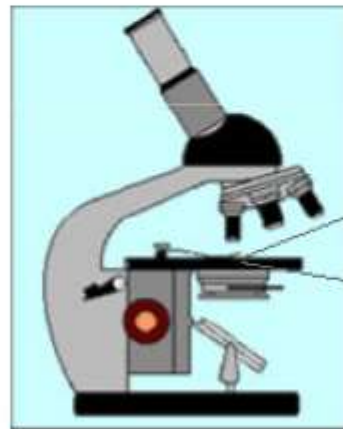
How can we get to nano?

New Tools!

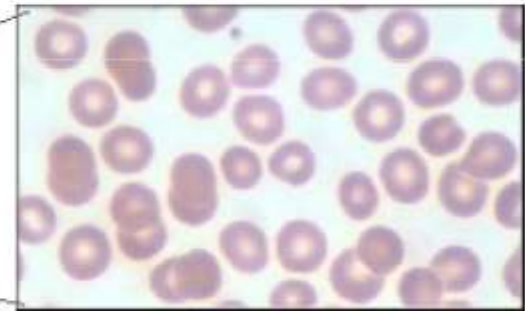
- **The naked eye can see to about 20 microns**
 - A human hair is about 50-100 microns thick
- **Light microscopes let us see to about 1 micron**
 - Bounce **light** off of surfaces to create images

Optical microscopes
use light to
see objects as small
as 200 nm.

! Invented in 1600s.



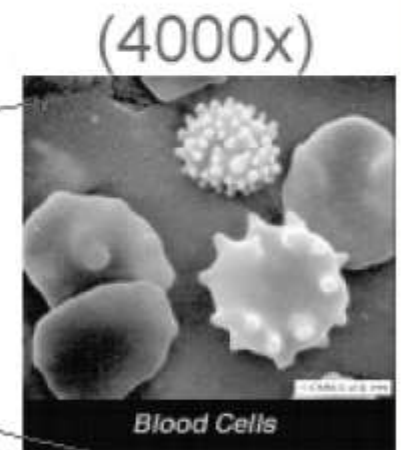
Light microscope
(magnification up to 1000x)



to see red blood cells
(400x)

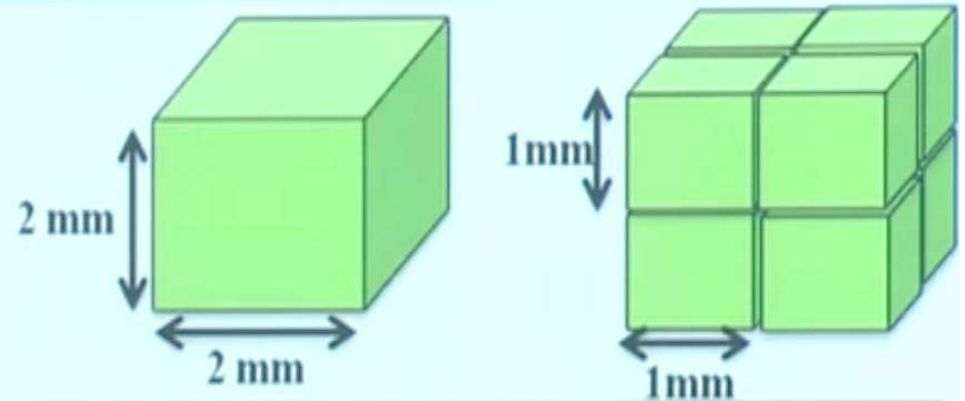
Electrons help to see small sized objects

- **Scanning electron microscopes (SEMs), invented in the 1930s, let us see objects as small as 10 nanometers**
 - Bounce electrons off of surfaces to create images
 - Higher resolution due to small size of electrons



Greater resolution to see things like blood cells in greater detail

Surface area to volume ratio



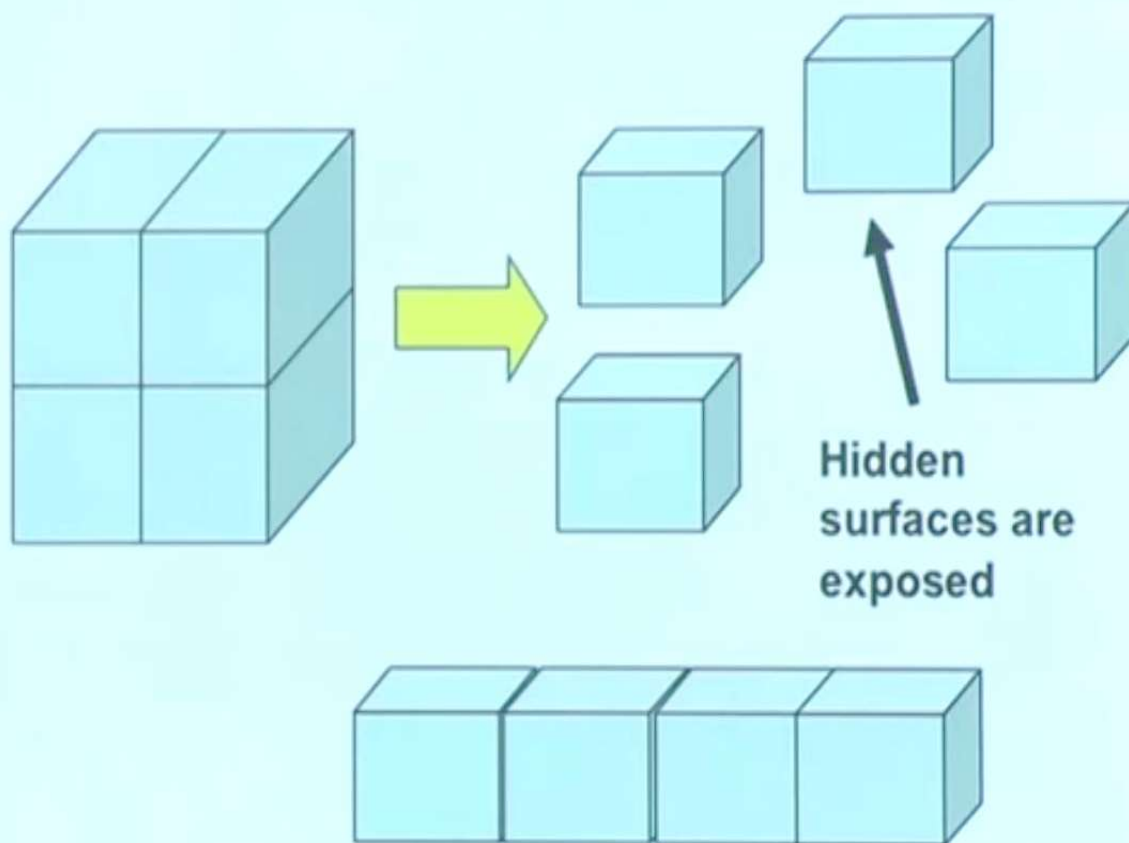
Surface Area (mm)	Surface area= Height x Width x No. of sides x No. of cubes	24 (2x2x6x1)	48 (1x1x6x8)
Volume (mm)	Volume=Height x Width x Length x No. of cubes	8 (2x2x2x1)	8 (1x1x1x8)
Surface Area/Volume ratio	Surface area/Volume	3 (24:8)	6 (48:8)

Surface area to volume ratio

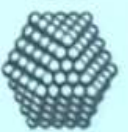
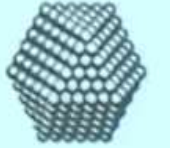
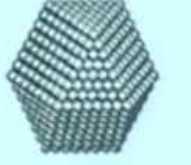
As surface to volume ratio increases

A greater amount of a substance comes in contact with surrounding material

This results in better catalysts, since a greater proportion of the material is exposed for potential reaction



Why small is good?

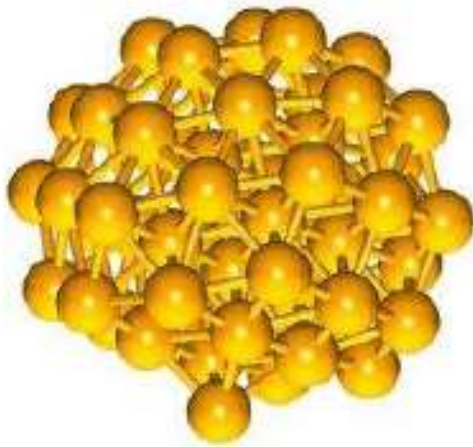
Full-shell Clusters		Total Number of Atoms	Surface Atoms (%)
1 Shell		13	92
2 Shells		55	76
3 Shells		147	63
4 Shells		309	52
5 Shells		561	45
7 Shells		1415	35

Nano-objects are:

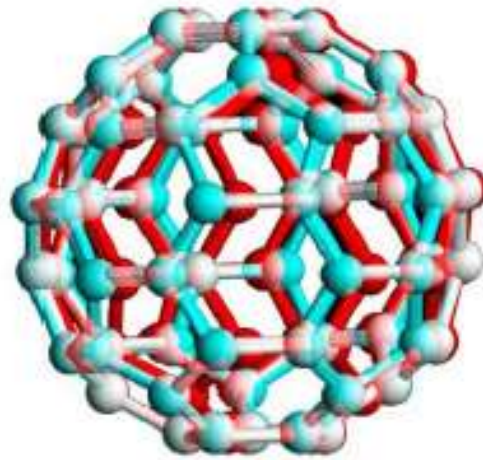
- Faster
- Lighter
- Can get into small spaces
- Cheaper
- More energy efficient
- Different properties at very small scale

Surface area increases as size decreases

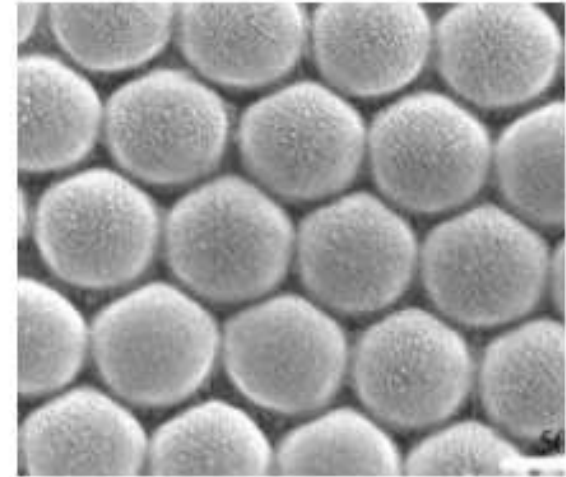
Examples of Nanomaterials



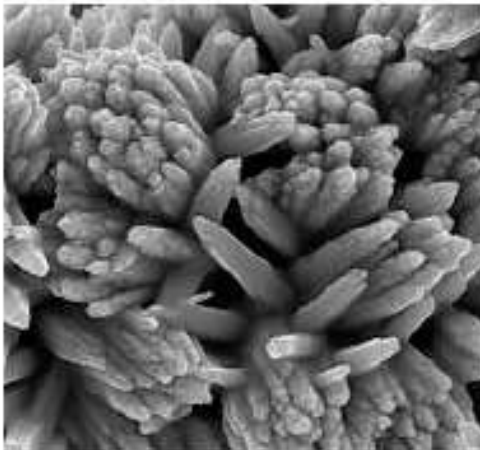
(a) Au nanoparticles



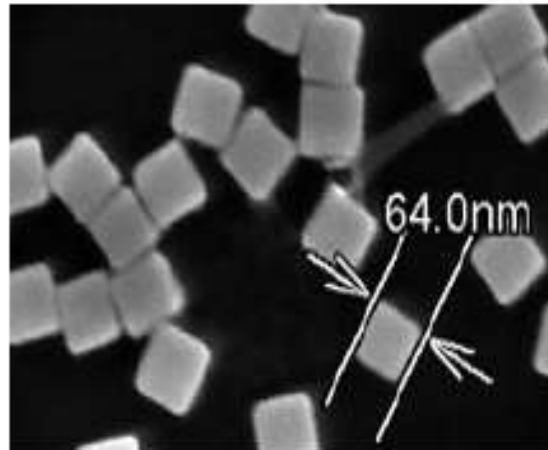
(b) Buckminsterfullerene



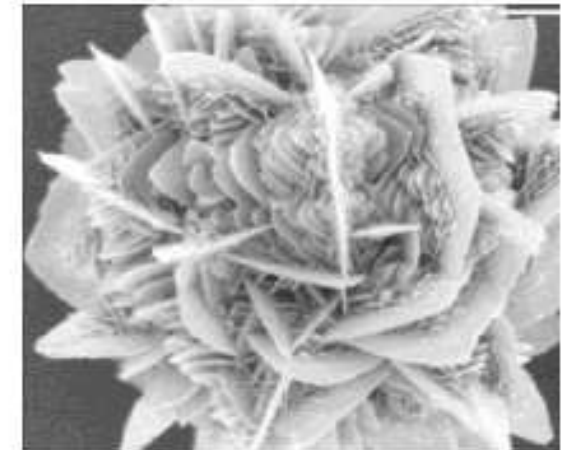
(c) FePt nanosphere



(d) Titanium nanoflower



(f) Titanium nanoflower

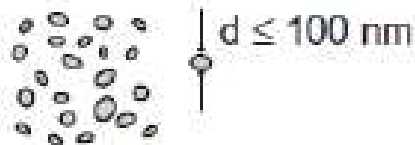


(g) SnO₂ nanoflower

- Classification is based on the number of dimensions, which are not confined to the nanoscale range (<100 nm).
- (1) zero-dimensional (0-D),
- (2) one-dimensional (1-D),
- (3) two-dimensional (2-D), and
- (4) three-dimensional (3-D).

0-D

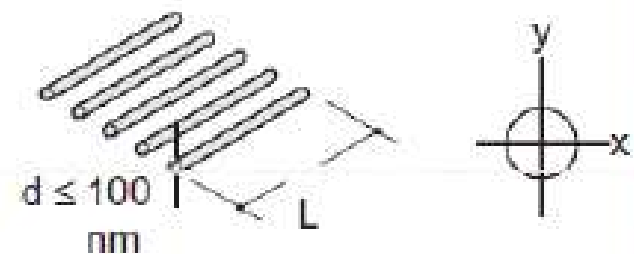
All dimensions (x, y, z) at nanoscale



Nanoparticles

1-D

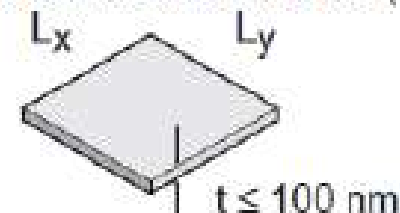
Two dimensions (x, y) at nanoscale, other dimension (L) is not



Nanowires, nanorods, and nanotubes

2-D

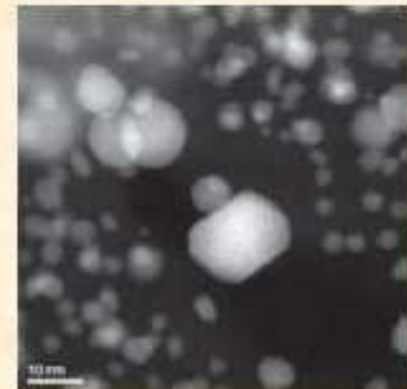
One dimension (t) at nanoscale, other two dimensions- (L_x, L_y) are not



Nanocoatings and nanofilms

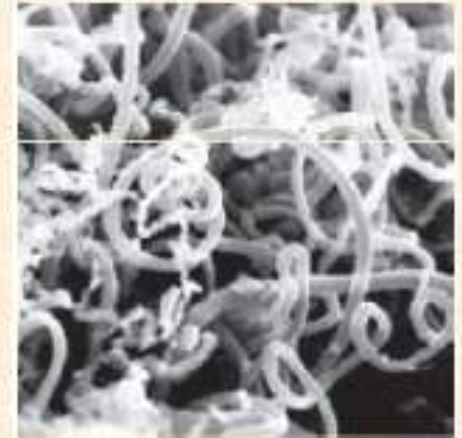
Zero Dimensional Nanomaterials

- Materials wherein all the dimensions are measured within the nanoscale (no dimensions, or 0-D, are larger than 100 nm).
- The most common representation of zero-dimensional nanomaterials are **nanoparticles**.
 - **Nanoparticles** can:
 - Be amorphous or crystalline
 - Be single crystalline or polycrystalline
 - Be composed of single or multi-chemical elements
 - Exhibit various shapes and forms
 - Exist individually or incorporated in a matrix
 - Be metallic, ceramic, or polymeric



One Dimensional Nanomaterials

- One dimension that is outside the nanoscale.
- This leads to needle like-shaped nanomaterials.
- 1-D materials include **nanotubes, nanorods, and nanowires**.
- **1-D nanomaterials** can be
 - Amorphous or crystalline
 - Single crystalline or polycrystalline
 - Chemically pure or impure
 - Standalone materials or embedded in within another medium
 - Metallic, ceramic, or polymeric

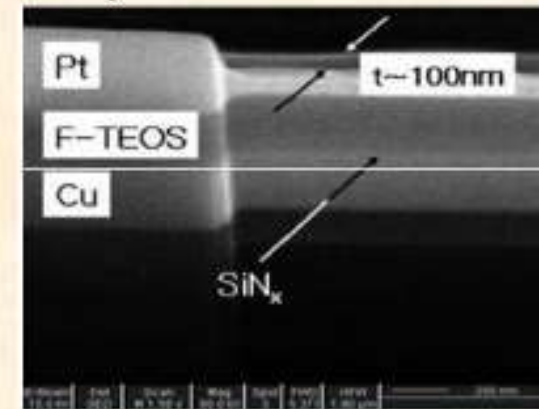


Two Dimensional Nanomaterials

- Two of the dimensions are not confined to the nanoscale.
- 2-D nanomaterials exhibit plate-like shapes.
- Two-dimensional nanomaterials include **nanofilms**, **nanolayers**, and **nanocoatings**.

- **2-D nanomaterials** can be:

- Amorphous or crystalline
- Made up of various chemical compositions
- Used as a single layer or as multilayer structures
- Deposited on a substrate
- Integrated in a surrounding matrix material
- Metallic, ceramic, or polymeric

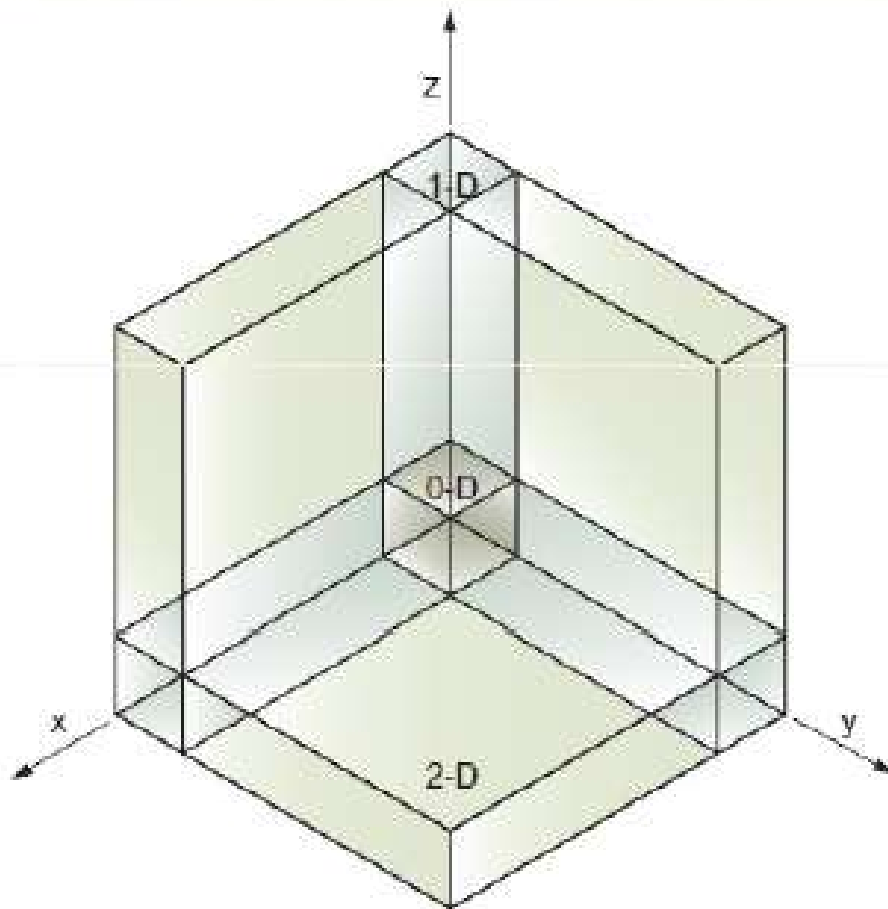




Three-dimensional nanomaterials

- **Bulk** nanomaterials are materials that are not confined to the nanoscale in any dimension. These materials are thus characterized by having three arbitrarily dimensions above 100 nm.
- Materials possess a nanocrystalline structure or involve the presence of features at the nanoscale.
- In terms of nanocrystalline structure, bulk nanomaterials can be composed of a multiple **arrangement of nanosize crystals**, most typically in different orientations.
- With respect to the presence of features at the nanoscale, **3-D nanomaterials** can contain dispersions of nanoparticles, bundles of nanowires, and nanotubes as well as multilayers.

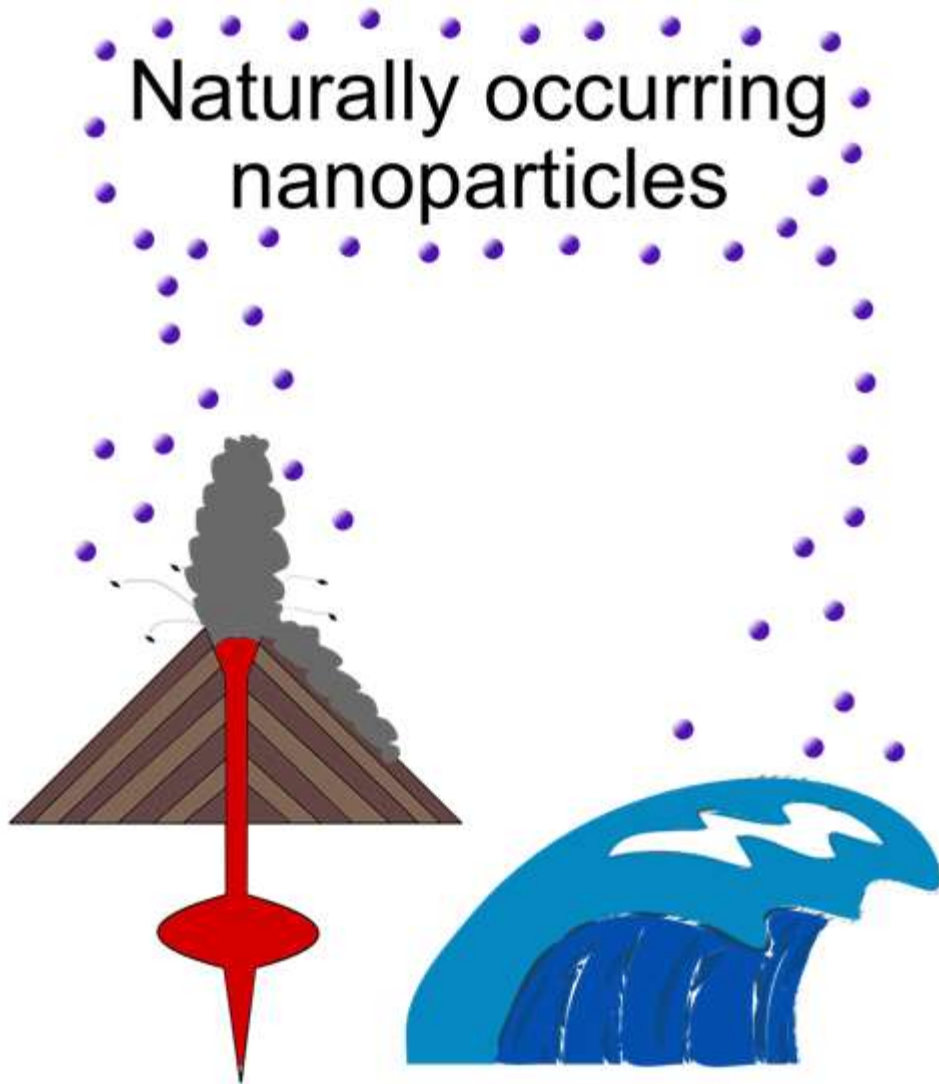
Three-dimensional space showing the relationships among 0-D, 1-D, 2-D, and 3-D nanomaterials



- 0-D: All dimensions at the nanoscale
- 1-D: Two dimensions at the nanoscale, one dimension at the macroscale
- 2-D: One dimension at the nanoscale, two dimensions at the macroscale
- 3-D: No dimensions at the nanoscale, all dimensions at the macroscale

Classification of nanomaterials based on their origin

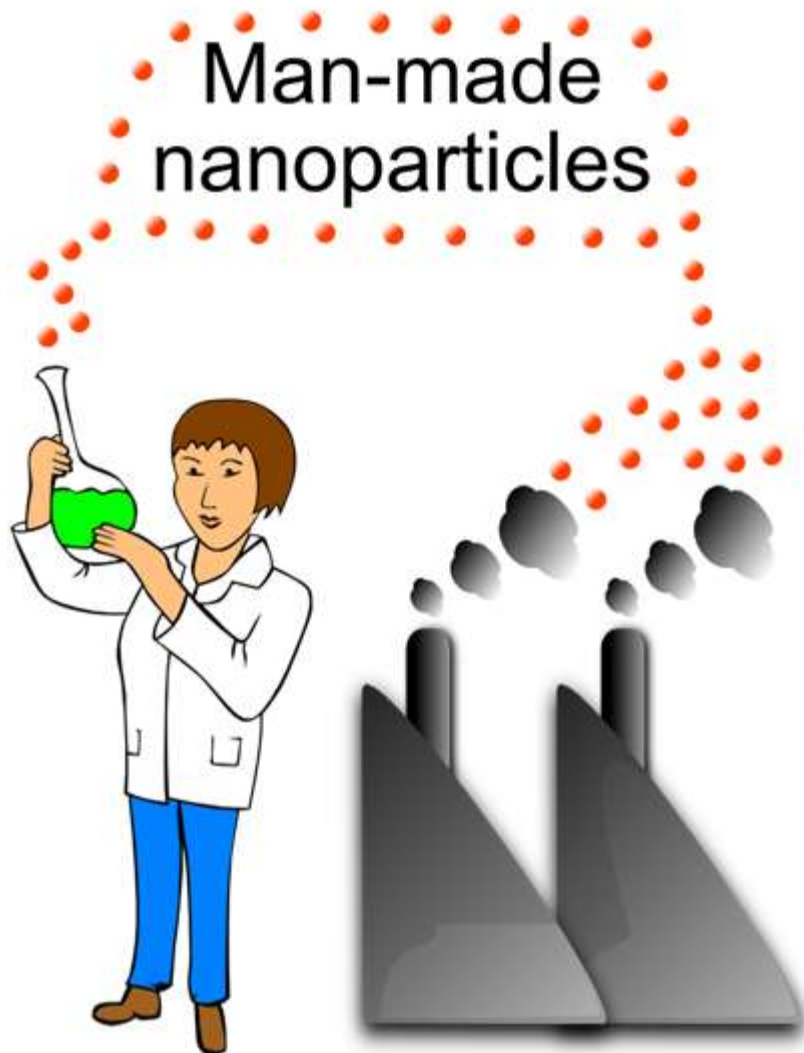
Naturally occurring nanoparticles



- ❖ Forest fire
- ❖ Volcanic ash
- ❖ Mineral composites
- ❖ Viruses

Source: <http://sustainable-nano.com/2013/03/25/nanoparticles-are-all-around-us/>

Classification of nanomaterials based on their origin



Human origin (Incidental):

- ❖ Diesel exhaust
- ❖ Welding fumes
- ❖ Industrial effluents
- ❖ Sandblasting
- ❖ Cooking smoke

Incidental Nanoparticles

more variable

different sizes
& shapes

composed of
different elements

100 nm

Engineered Nanoparticles

more well-defined

"shell" material

"core" material

organic
coating

100 nm

Human origin (Engineered):

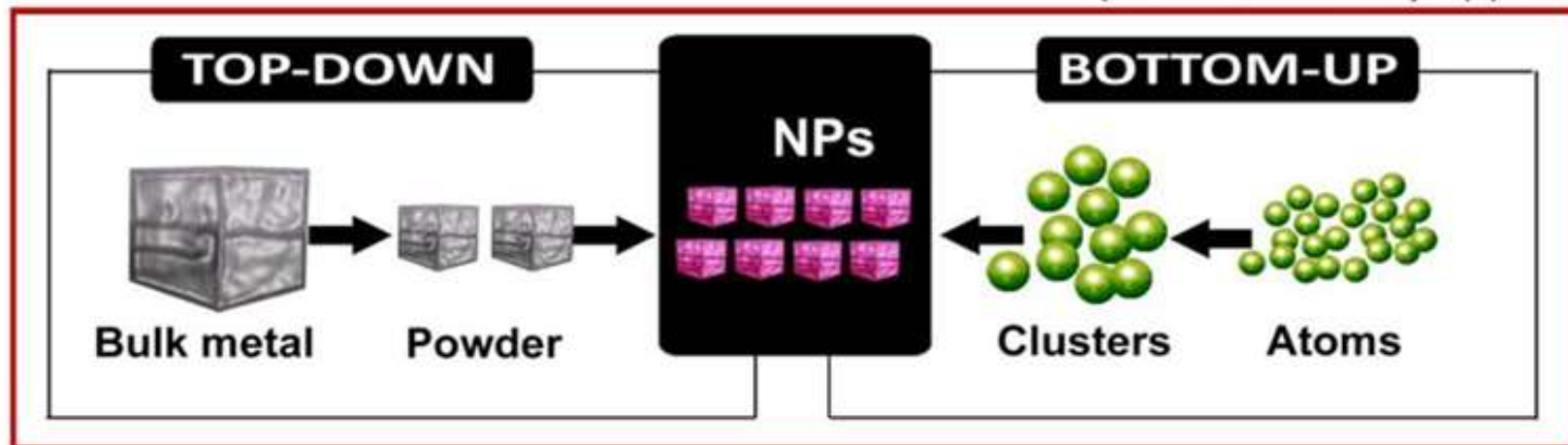
- ❖ Metals
- ❖ Quantum dots
- ❖ Nanotubes
- ❖ Ceramics
- ❖ Polymer
- ❖ Sunscreen pigments

How to make Nanoparticles

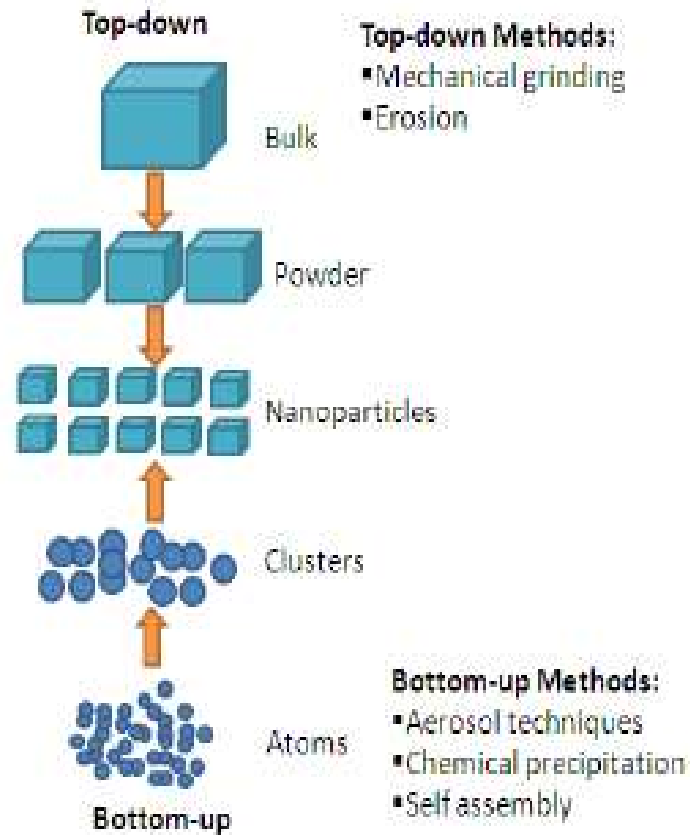
There are two approaches for synthesis of nanomaterials and the fabrication of nano structures.

Top-Down approach

Bottom-Up approach
(or self-assembly approach)



Synthesis Approaches and Techniques



Top-down approach refers to slicing or successive cutting of a bulk material to get nanosized particles

Ex: Lithography, Ball milling, Epitaxy, etc.

Bottom-up approach refers to the build-up of a material from the bottom: atom-by-atom, molecule-by-molecule, or cluster-by-cluster.

Ex: Sol-Gel Process, Chemical Reduction methods, etc.

Different nano-synthesis methods

Physical

- Ball milling
- Thermal evaporation
- Lithography
- Vapour phase

Chemical

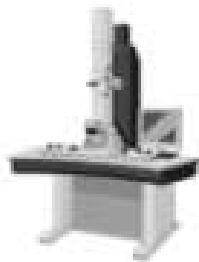
- Sol-gel processing (films)
- Solution-based synthesis

Biological

- In bacteria
- In fungi
- Using plant extracts

In water
(At the air-water
interface and in foams !)

In non-polar
organic solvents



Electron microscopy
Scanning Electron Microscopy (SEM)
Environmental Scanning Electron Microscopy (ESEM)
Transmission electron microscopy (TEM)

Neutron and X-ray Diffraction (XRD)
Circular Dichroism (CD)
Dynamic light scattering (DLS)

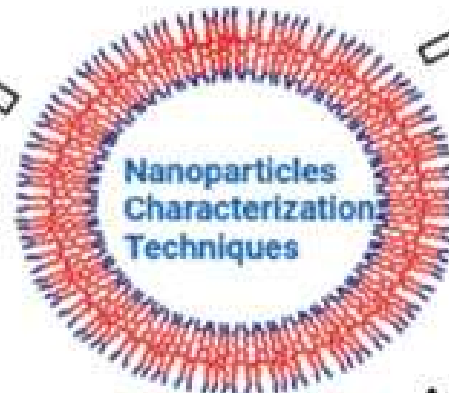


Optical measurement
Zeta potential (ZP)

Scanning probe microscopy
Atomic Force Microscope (AFM)
Scanning Tunneling Microscope (STM)



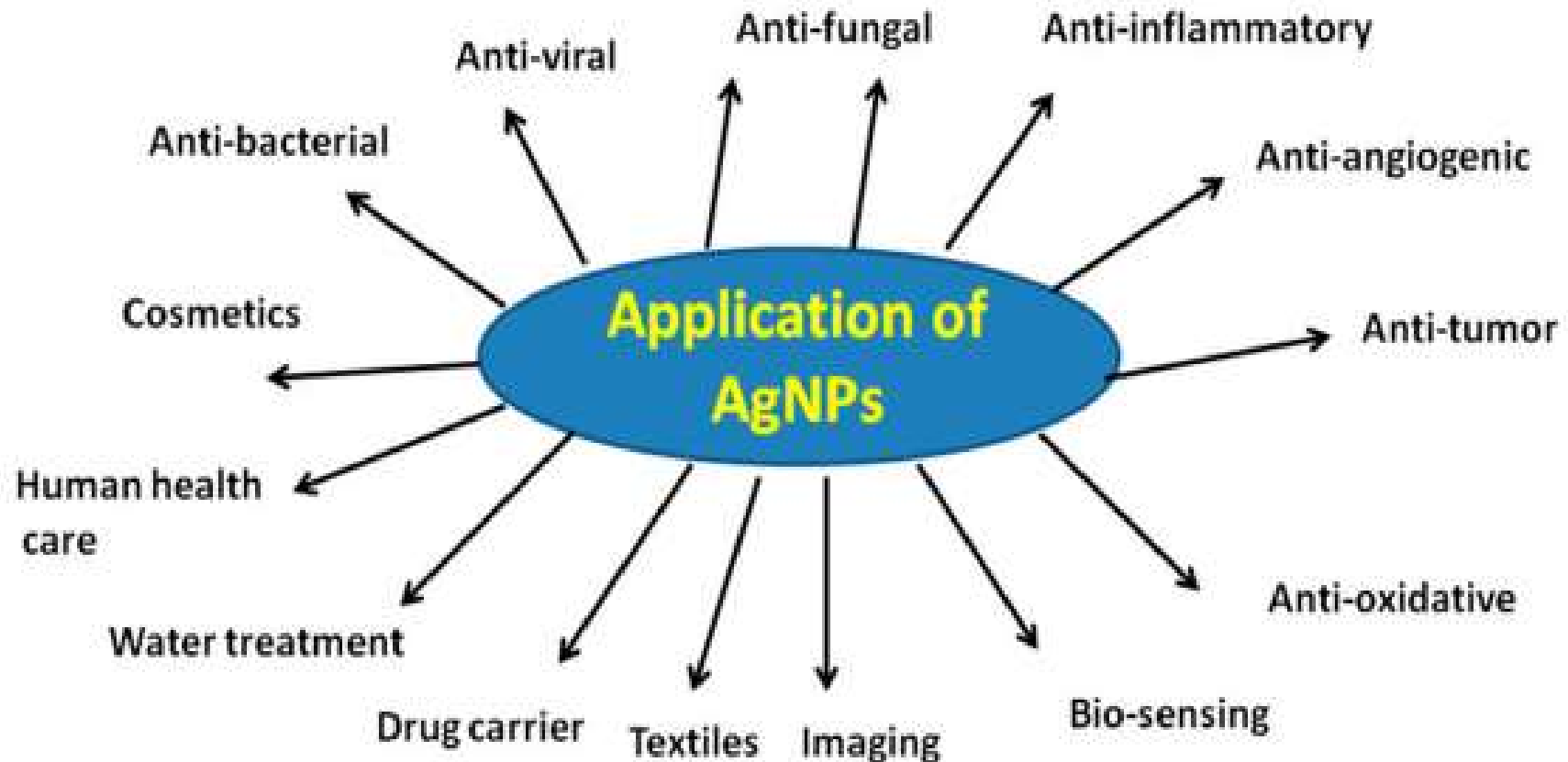
Complementary Analytical Techniques
Mass Spectroscopy (MS)
Surface-Enhanced Raman Spectroscopy (SERS)
Tip-Enhanced Raman Spectroscopy (TERS)



Analytical Techniques
Ultraviolet-Visible Spectroscopy Studies (UV-Vis)
Fourier Transforms Infrared Spectroscopy (FTIR)
Fluorescence spectroscopy (FS)
Nuclear magnetic resonance spectroscopy (NMR)

Nanoparticles characterization techniques.

Applications of AgNPs





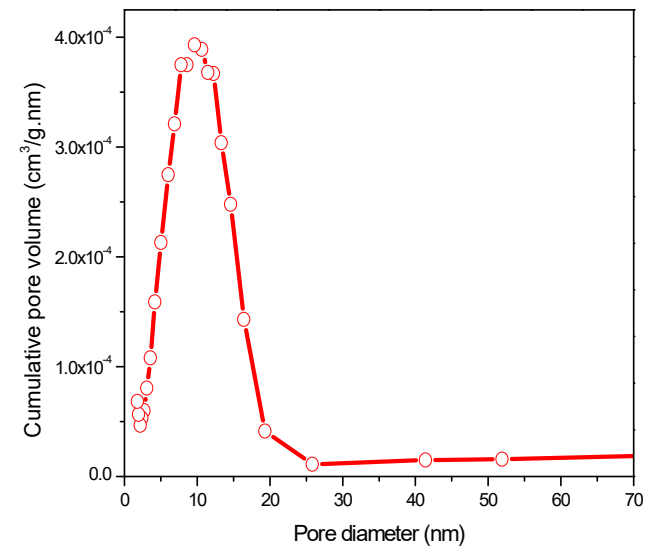
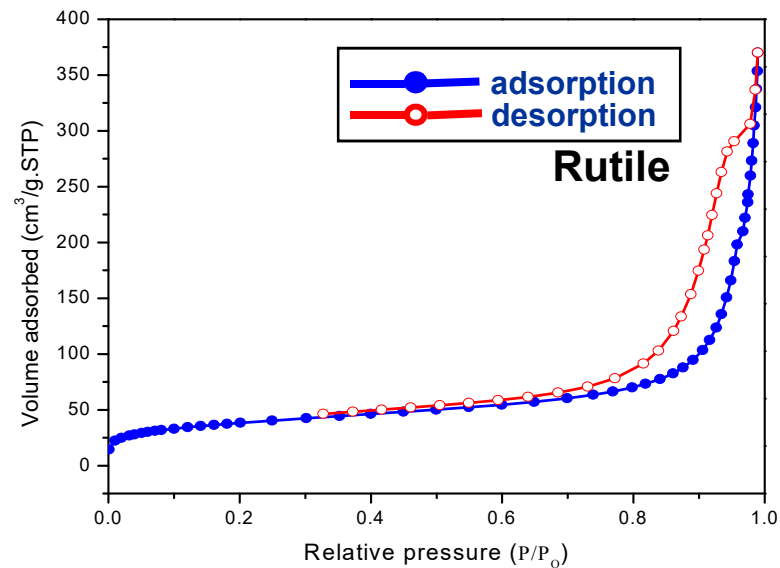
Applications of Nanoparticles

- a) **Silver nanoparticles** have good antibacterial properties, and are used in surgical instruments, refrigerators, air-conditioners, water purifiers etc.
- b) **Gold nanoparticles** are used in catalytic synthesis of silicon nano wires, sensors carrying the drugs and in the detection of tumors.
- c) **ZnO nanoparticles** are used in electronics, ultraviolet (UV) light emitters, piezoelectric devices and chemical sensors.
- d) **TiO₂ nanoparticles** are used as photocatalyst and sunscreen cosmetics (UV blocking pigment).
- e) **Antimony-Tin-Oxide (ATO), Indium-Tin-Oxide (ITO)** nanoparticles are used in car windows, liquid crystal displays and in solar cell preparations.

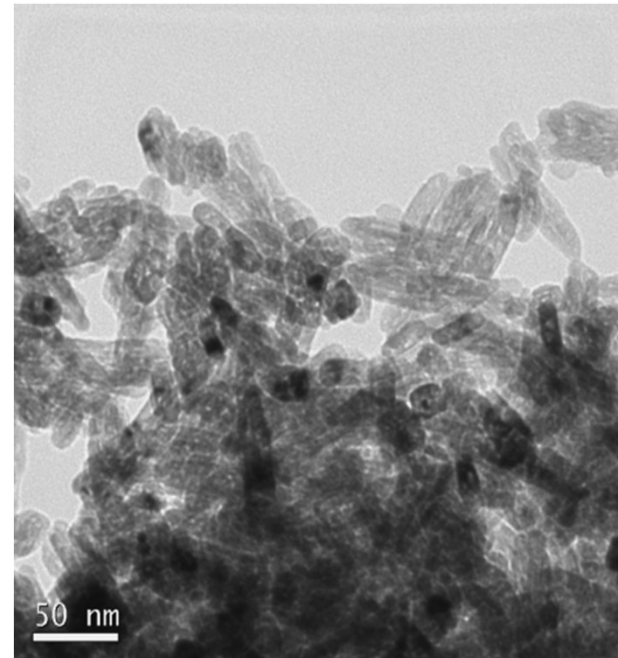
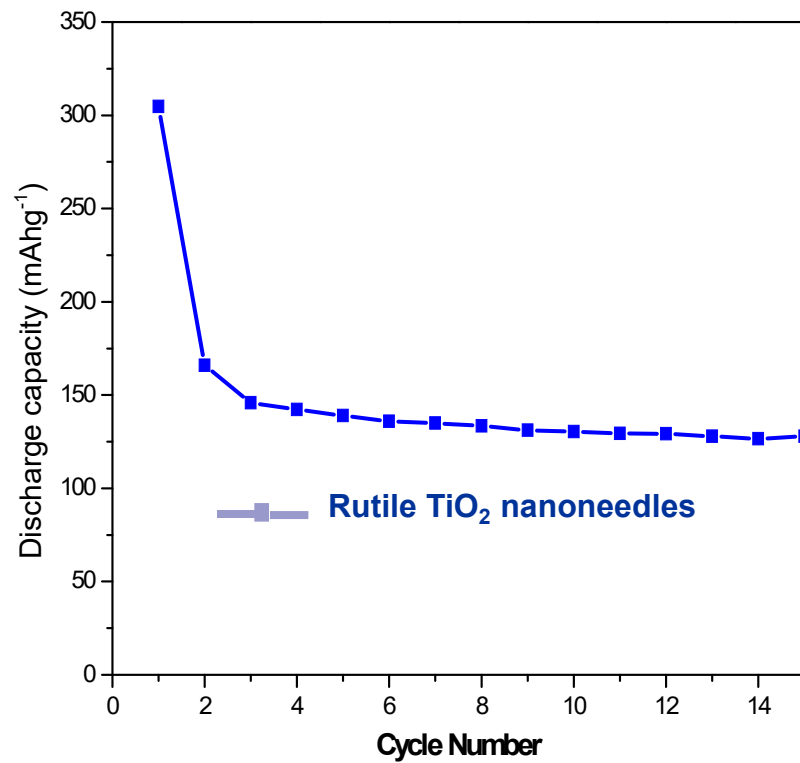


Nanomaterials for Energy Storage in Lithium-ion Battery Applications

Rutile TiO₂ Nanoneedles



LIB APPLICATION



Rutile TiO₂ nanoneedles

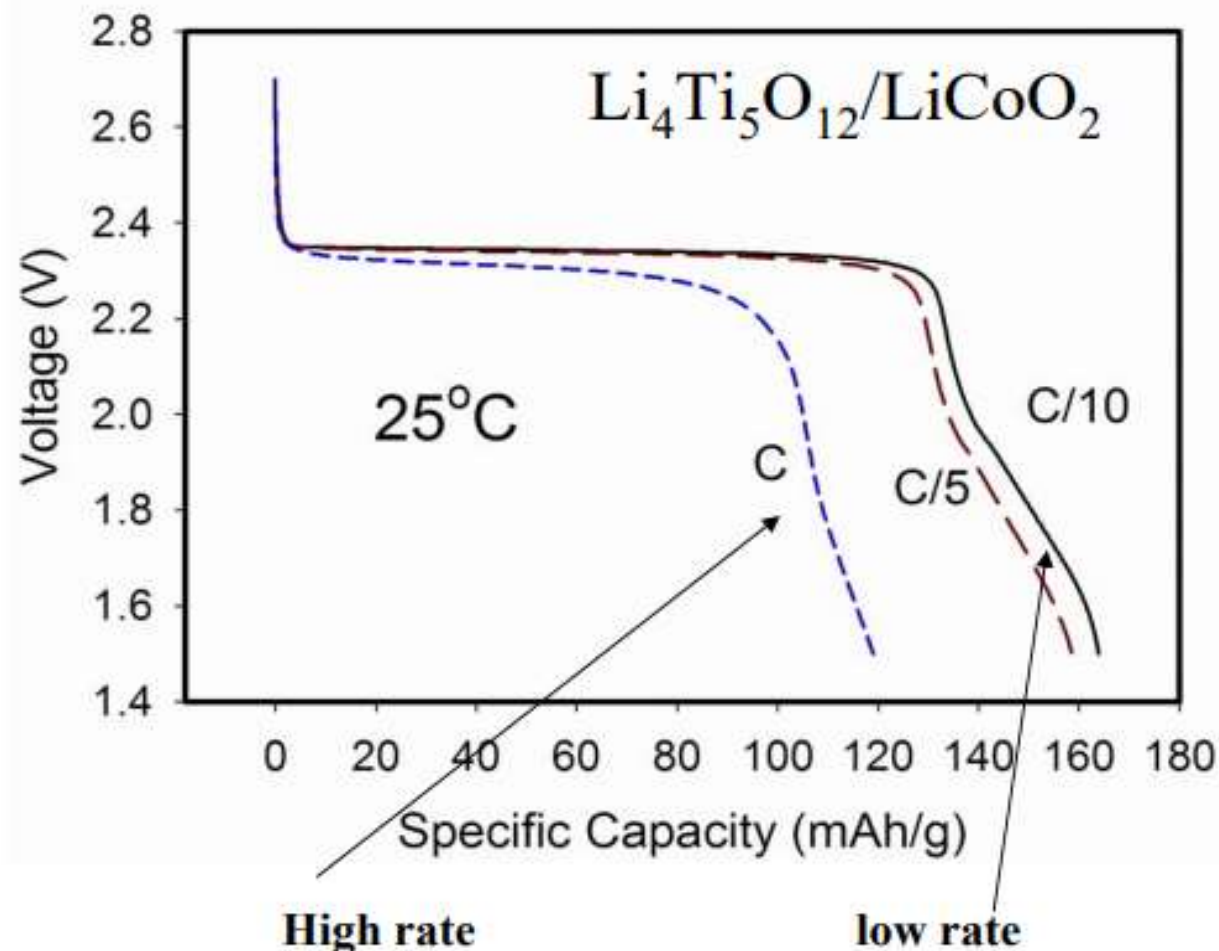
Lithium Ion Batteries

- $\text{Li}_4\text{Ti}_5\text{O}_{12}$
- LiCoO_2

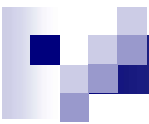


R.B.Khomane et al. Mater Res. Bulletin 46 (2011) 1139.
R.B. Khomane et al. Mater Res. Bulletin 43(2008) 2497.

Nanoparticles can improve rate capability



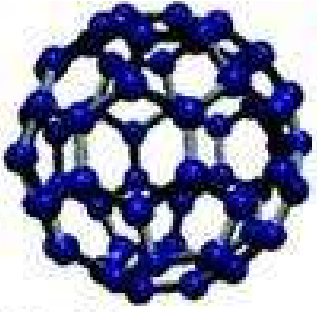
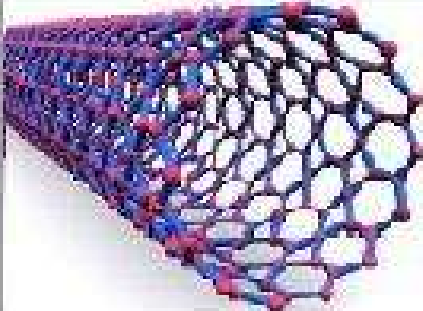
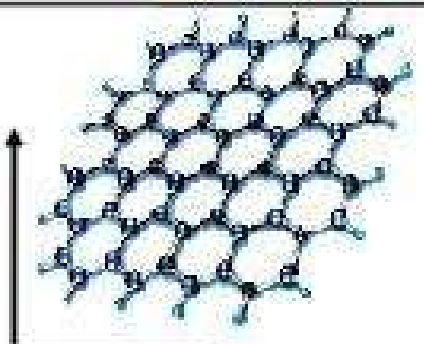
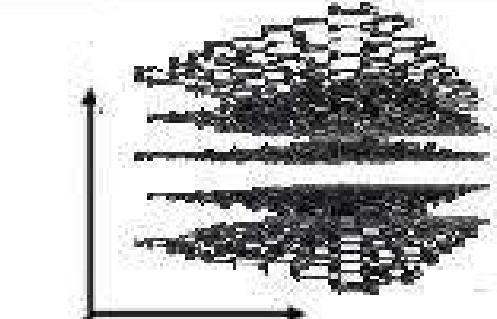
Charge and discharge rates of a battery are governed by C-rates. ... A C-rate of 1C is also known as a one-hour discharge; 0.5C or C/2 is a two-hour discharge and 0.2C or C/5 is a 5-hour discharge.



Advantages of NM

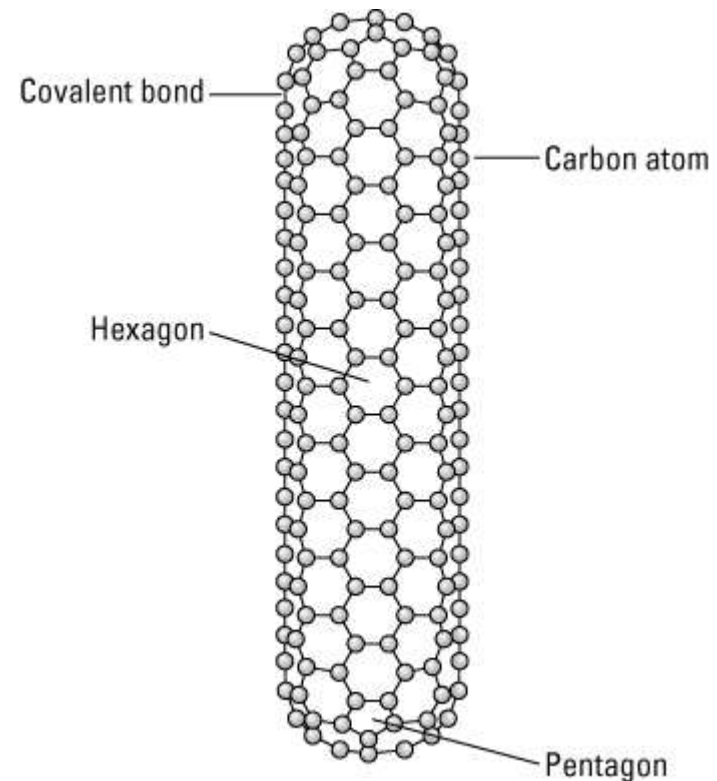
- Small particle size decreases electron diffusion parameters (high rate capability). Shorter diffusion lengths for the Lithium-ion travelling from the particle core to the surface where it transfers to the electrolyte.
- High surface area allows active material to absorb lithium ions more effectively (high capacity).
- Small particle size may accommodate crystalline expansion of lattice (improved cyclability).

Carbon Nanostructures

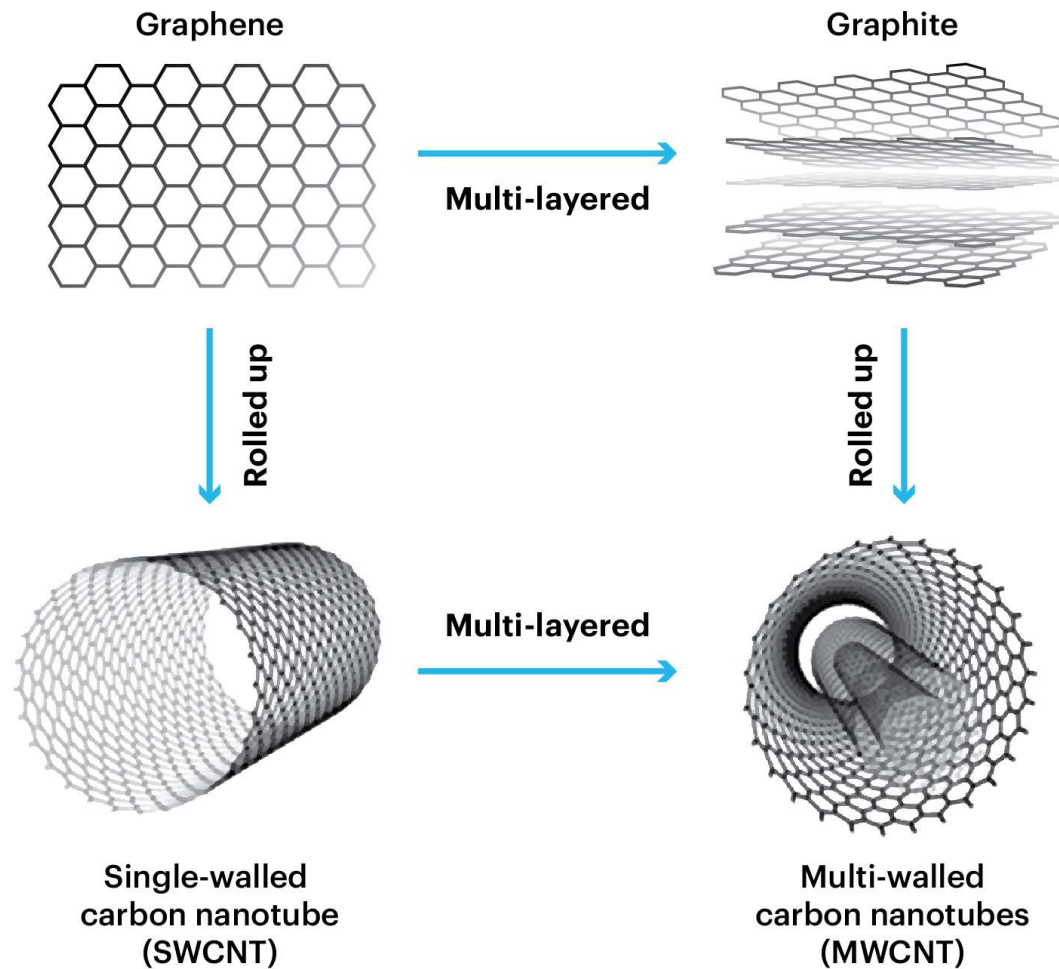
0D	1D	2D	3D
			
Fullerene	Carbon Nanotube	Graphene	Graphite

CARBON NANOTUBES (CNTs)

- ▶ *A Carbon Nanotube is a tube-shaped material, made of carbon, having a diameter measuring on the nanometer scale.*
- ▶ *The graphite layer appears somewhat like a rolled-up with a continuous unbroken hexagonal mesh and carbon molecules at the apexes of the hexagons. Their name is derived from their long, hollow structure with the walls formed by one-atom thick sheets of carbon, called graphene.*



CARBON NANOTUBES (CNTs)



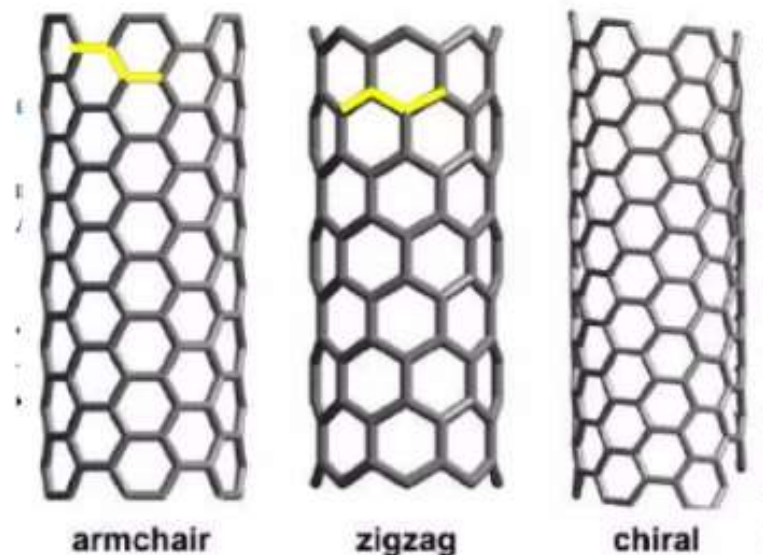


Types of Carbon Nanotubes

1. *Single Walled CNTs*
2. *Multi Walled CNTs*

Single-walled carbon nanotube structure

- Single-walled carbon nanotubes can be formed in three different designs: Armchair, Chiral, and Zigzag.
- The design depends on the way the graphene is wrapped into a cylinder.



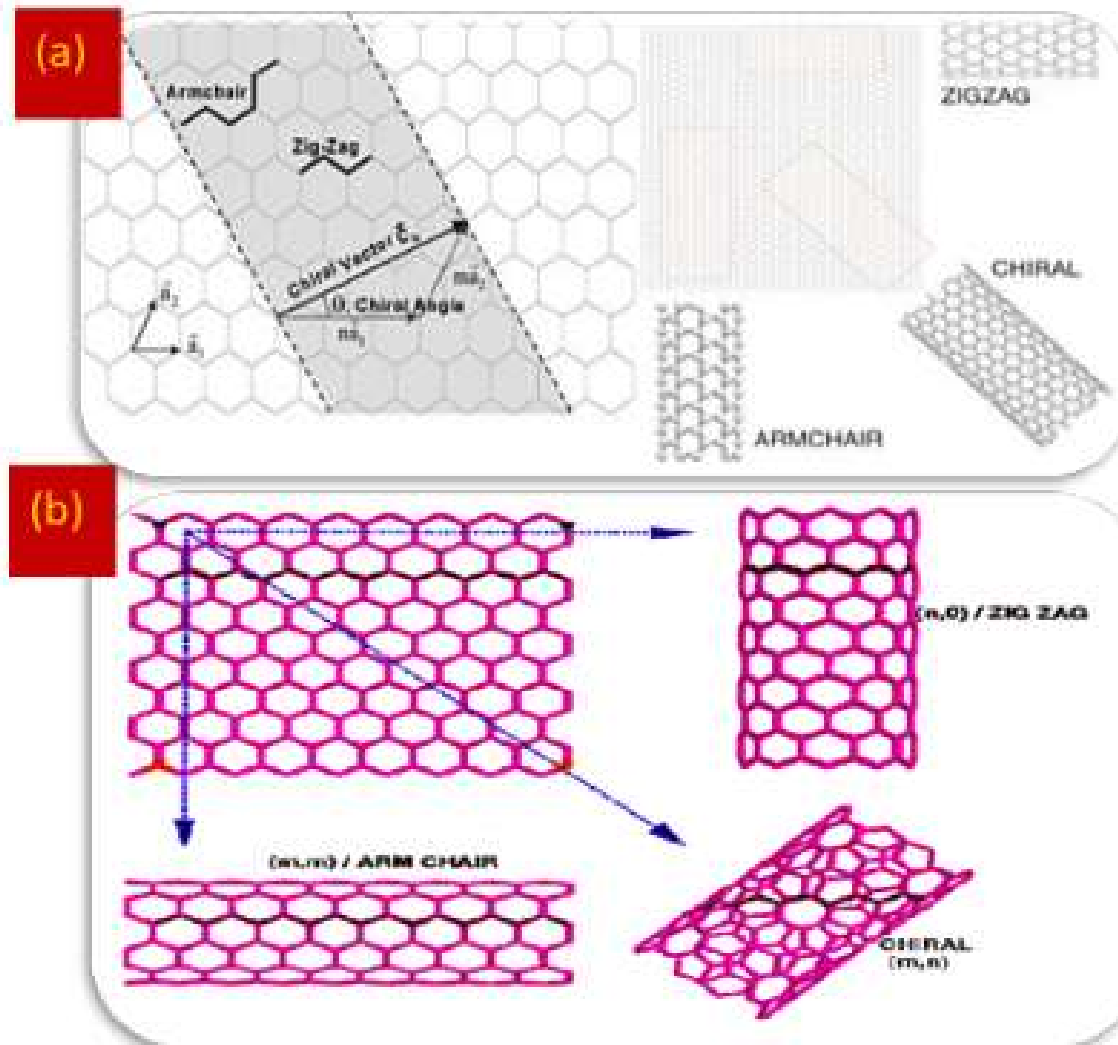
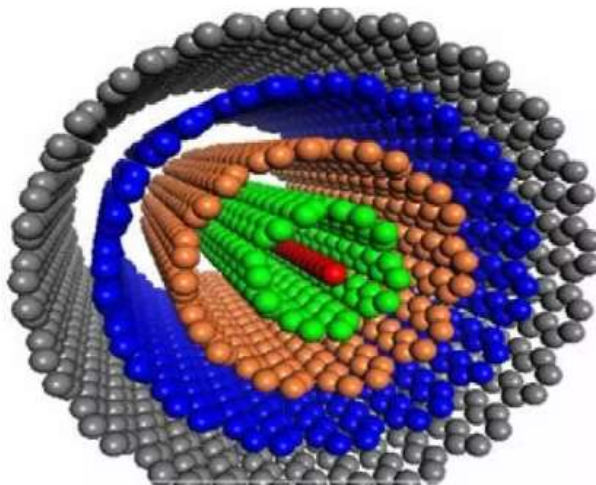


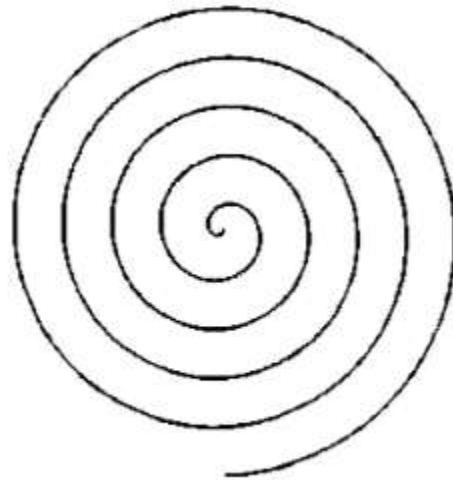
Figure 4: (a) Various parameters of CNTs (b) Helicity of graphene sheet.

Multi-walled carbon nanotube structure

- There are two structural models of multi-walled nanotubes:
 - Russian Doll model
 - Parchment model
- In the **Russian Doll model**, a carbon nanotube contains another nanotube inside it (the inner nanotube has a smaller diameter than the outer nanotube).



- In the **Parchment model**, a single graphene sheet is rolled around itself multiple times, resembling a rolled up scroll of paper.

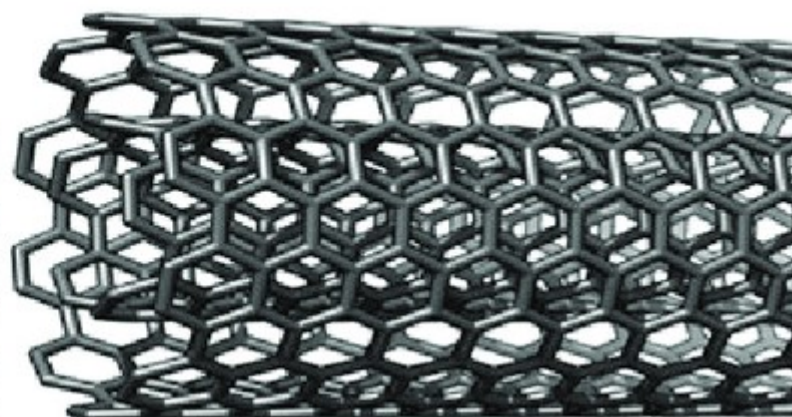




Single-walled carbon nanotubes (SWCNTs)



Multi-walled carbon nanotubes (MWCNTs)



It is a single layer of graphene

It requires catalyst for synthesis

Bulk synthesis is difficult

More defection during functionalization

Poor in purity

Less accumulation in the body

It can twist easily

Easy characterization and evaluation

It is a multiple layer of graphene

It can be produced without catalyst

Bulk synthesis is easy

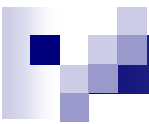
Less defection, but difficult to improve

Purity is high

More accumulation in the body

It is difficult to twist

Difficult characterization and evaluation

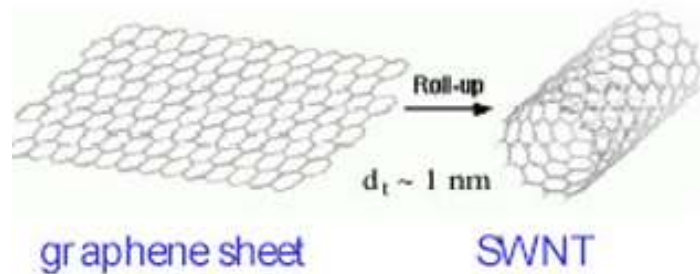


Properties of nanotubes

- Carbon nanotubes are long meshed wires of carbon
- Longest tubes up to 1mm long and few nanometers thick made by IBM.

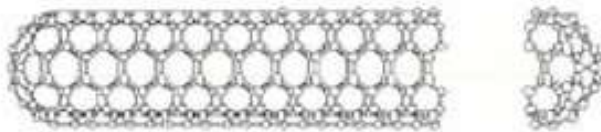
Property	Carbon Nanotubes	Comparatively
Size	0.6-1.8 nm in diameter	Si wires at least 50nm thick
Strength	45 Billion Pascals	Steel alloys have 2 Billion P.
Resilience	Bent and straightened without damage	Metals fracture when bent and restraightened
Conductivity	Estimated at 10^9 A/cm ²	Cu wires burn at 10^6 A/cm ²
Cost	\$2500/gram by BuckyUSA in Houston	Gold is \$15/gram

Overview of NT properties



- **Size:** Nanostructures with dimensions of $\sim 1 \text{ nm}$ diameter (~ 10 atoms around the cylinder)

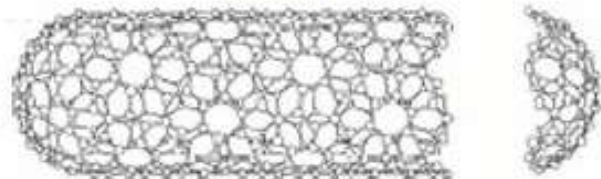
armchair



zigzag



chiral



- **Electronic Properties:** Can be either metallic or semiconducting depending on diameter and orientation of the hexagons
- **Mechanical:** Very high strength, modulus, and resiliency. Good properties on both compression and extension.
- **Physics:** 1D density of electronic states
- Single molecule Raman spectroscopy and luminescence.
- Single molecule transport properties.
- Heat pipe, electromagnetic waveguide.



Applications of CNTs

- Wind Mill Blades:** Nanotubes are also used in the wind mill blades because of their low weight to better wind and produce more electricity at faster rate.

Filtration: CNT can be used to separate particles of size greater than diameter of CNT, during filtration through CNT. CNT can also be used to trap smaller sized ions from a solution.

CNT as Nanocylinders: A gas like H₂, for energy, battery or vehicles can be safely stored inside the CNT and problem of H₂ storage-hazards can be solved. CNTs have also been shown to absorb Infrared light and may have applications in the I/R Optics Industry



Air Craft Stress Reduction: Nanotubes are also used in the space and air craft's to reduce the weight and stress of the various components working tighter.

Catalyst in some reactions

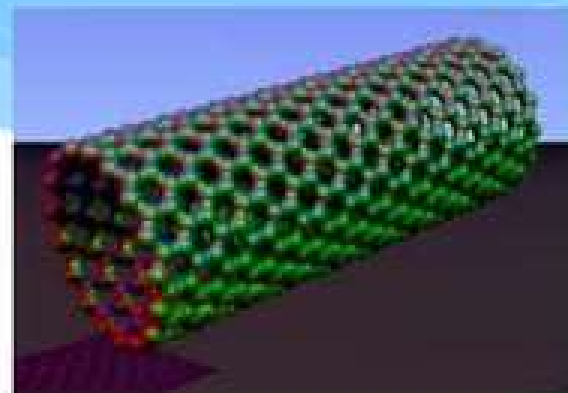
Used in drug delivery system

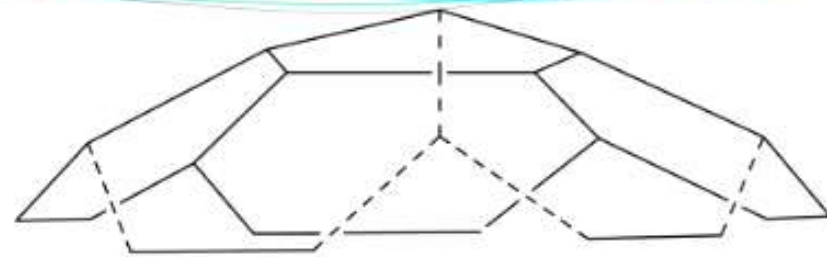
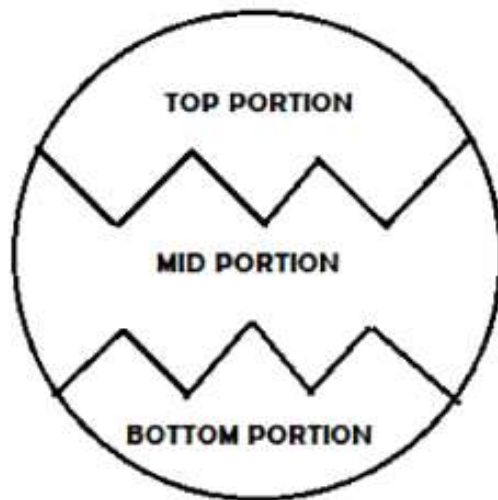
Applications related to conductivity, in electronics

FULLERENE

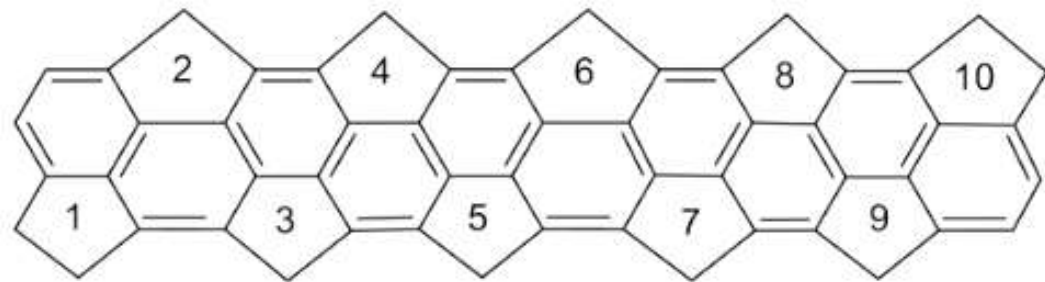
FULLERENE INTRODUCTION

A **Fullerene** is any molecule composed entirely of carbon, in the form of a hollow **sphere**, **ellipsoid**, or **tube**. Spherical fullerenes are also called **buckyballs**, and cylindrical ones are called **carbon nanotubes** or **buckytubes**.

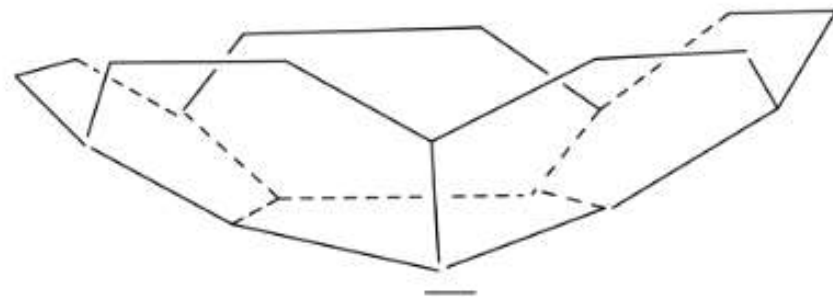




A. Top portion (1 pentagon, 5 hexagons)



B. Midportion (10 hexagons, 10 pentagons)



C. Bottom portion(1 pentagon, 5 hexagons)

NAMING OF FULLERENE

Buckminsterfullerene (C_{60}) was named after Richard Buckminster Fuller, a noted architectural modeler who popularized the geodesic dome.

The shortened name 'fullerene' is used to refer to the family of fullerenes. The suffix “ene” indicates that each C atom is covalently bonded to three others.

Buckminster Fuller



Richard Buckminster Fuller, c. 1917

Buckminster Fuller	
Richard Buckminster Fuller, c. 1917.	
Born	July 12, 1895 Milton, Massachusetts , United States
Died	July 1, 1983 (aged 87) Los Angeles , United States
Occupation	Visionary , designer , architect , author, inventor
Spouse	Anne Fuller
Children	2: Allegra Fuller Snyder and Alexandra who died in childhood



The [Montreal Biosphère](#) by Buckminster Fuller, 1967

TYPES OF FULLERENE

BUCKYBALL CLUSTERS: smallest member is C_{20} (unsaturated version of dodecahedrane) and the most common is C_{60} .

NANOTUBES: hollow tubes of very small dimensions, having single or multiple walls; potential applications in electronics industry.

MEGATUBES: larger in diameter than nanotubes and prepared with walls of different thickness; potentially used for the transport of a variety of molecules of different sizes.

POLYMERS: chain, two-dimensional and three-dimensional polymers are formed under high-pressure high-temperature conditions; single-strand polymers are formed using the Atom Transfer Radical Addition Polymerization (ATRAP) route.

NANO"ONIONS": spherical particles based on multiple carbon layers surrounding a buckyball core; proposed for lubricants.

LINKED "BALL-AND-CHAIN" DIMERS: two buckyballs linked by a carbon chain.



PROPERTIES OF FULLERENE

AROMATICITY

Researchers have been able to increase the reactivity of fullerenes by attaching active groups to their surfaces.

Buckminsterfullerene does not exhibit "superaromaticity": that is, the electrons in the hexagonal rings do not delocalize over the whole molecule.

CHEMISTRY

Fullerenes are stable, but not totally unreactive.

The sp^3 -hybridized carbon atoms

Other atoms can be trapped inside fullerenes to form inclusion compounds known as *endohedral fullerenes*.

SOLUBILITY

Fullerenes are sparingly soluble in many solvents.

Common solvents for the fullerenes include aromatics, such as toluene, and others like carbon disulfide.

Solutions of pure buckminsterfullerene have a deep purple color. Solutions of C_{70} are a reddish brown.

Fullerenes are the only known allotrope of carbon that can be dissolved in common solvents at room temperature.

Quantum mechanics

In 1999, researchers from the University of Vienna demonstrated that wave-particle duality applied to molecules such as fullerene.

Science writer Marcus Chown stated on the CBC radio show Quirks and Quarks in May 2006 that scientists are trying to make buckyballs exhibit the quantum behavior of existing in two places at once.

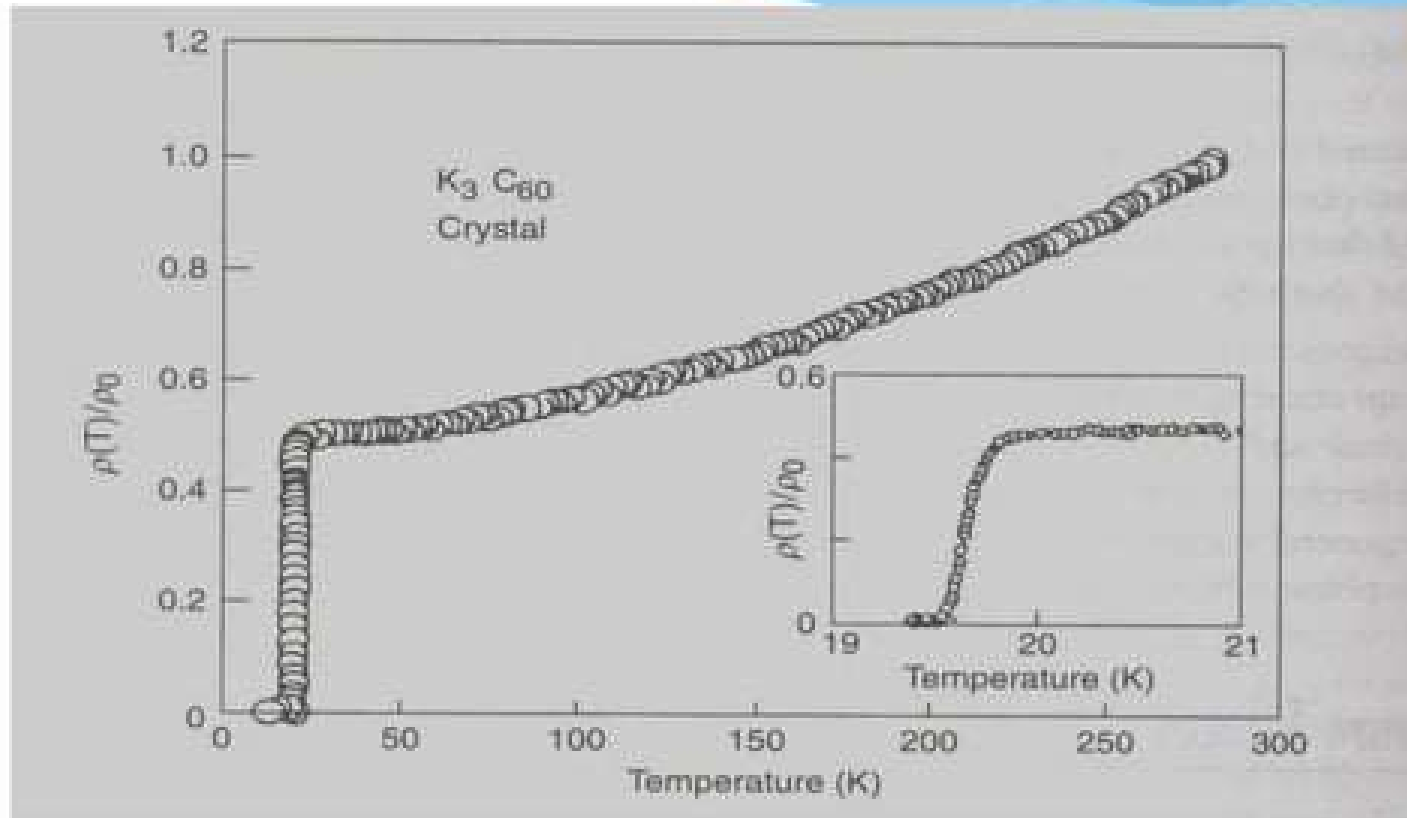
CHIRALITY

Some fullerenes (e.g. C₇₆, C₇₈, C₈₀, and C₈₄) are inherently chiral because they are D₂-symmetric, and have been successfully resolved. Research efforts are ongoing to develop specific sensors for their enantiomers

SUPERCONDUCTIVITY IN FULLERENE

In 1991, it was revealed that potassium-doped C₆₀ becomes superconducting at 18 K. This was the highest transition temperature for a molecular superconductor. Since then, superconductivity has been reported in fullerene doped with various other alkali metals

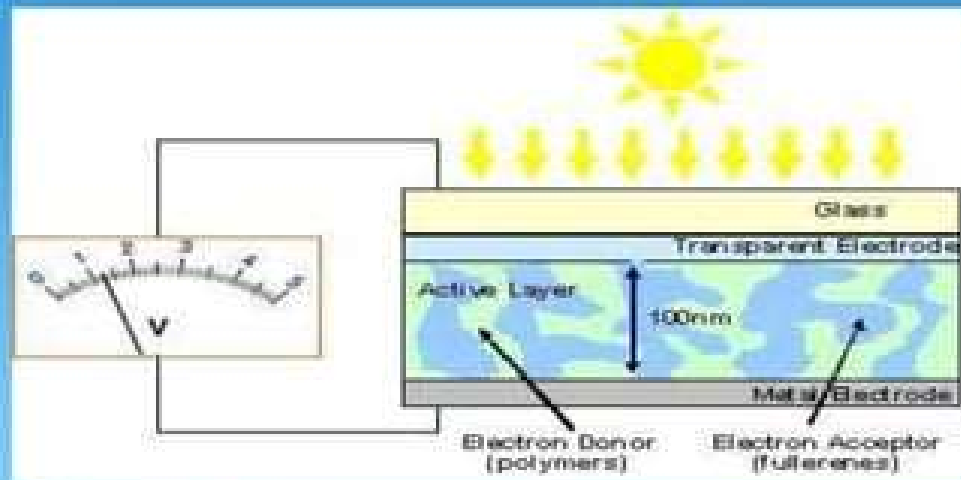
CONDUCTIVITY IN FULLERENE



Normalized DC electrical resistivity $\rho(T)$ of a $K_3 C_{60}$ single crystal. The T_c observed is 19.8K. ρ_0 is the resistivity at $T=280K$.

APPLICATIONS

1. Organic Photovoltaics (OPV)



2. Polymer Electronics

The performance of polymer transistors (Organic Field Effect Transistors (OFETS)) and photodetectors has also been increasing, in part due to a great deal of synergy between OFETS and OPVs

Antioxidants & Biopharmaceutical

Fullerenes are powerful antioxidants, reacting readily and at a high rate with free radicals, which are often the cause of cell damage or death. Fullerenes hold great promise in health and personal care applications where prevention of oxidative cell damage or death is desirable, as well as in non-physiological applications where oxidation and radical processes are destructive (food spoilage, plastics deterioration, metal corrosion).

Polymer Additives

Fullerenes and fullerenic black are chemically reactive and can be added to polymer structures to create new copolymers with specific physical and mechanical properties. They can also be added to make composites. Much work has been done on the use of fullerenes as polymer additives to modify physical properties and performance characteristics.

OTHER APPLICATIONS

Catalysts

Marked ability to accept and to transfer hydrogen atoms; hydrogenation and hydrodealkylations.

Highly effective in promoting the conversion of methane into higher hydrocarbons.
Inhibits coking reactions.

Water purification & bio-hazard protection

Singlet oxygen catalysis of organics with fullerene C₆₀.

Portable power

Proton exchange membranes for fuel cells.

Vehicles

Enhanced durability, lower heat build-up, better fuel economy with use of fullerene black/rubber compounds.

Medical

MRI agents.

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